



AMAN DEMROT

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GitHub Profile LinkedIn Profile
Malaviya National Institute of Technology, Jaipur | Roll No: 2023UCH1213

EDUCATION

Malaviya National Institute of Technology, Jaipur	2023 – 2027
B.Tech in Chemical Engineering	
RBSE Board (Class XII)	2022
Engineers Point, Alwar	Percentage: 83.5%
RBSE Board (Class X)	2020
SSDM School, Alwar	Percentage: 83.6%

EXPERIENCE

Web Development Intern	May 2026 – July 2026
Bridgestone Technologies	Remote
- Assisted in building lightweight backend services with Node.js and Express, implementing RESTful API endpoints for dynamic user data. - Designed and developed responsive user interfaces and dashboard components using React.js and CSS, ensuring seamless integration with backend APIs for dynamic data rendering. - Optimized web application performance by streamlining database queries and state management, resolving minor bottlenecks to improve overall page load times and user experience. - Optimized site performance and UI elements, leading to faster load times and improved user retention rates.	

PERSONAL PROJECTS

IntelliDocs – RAG-Based AI Knowledge Assistant | [GitHub](#) | [Live](#)

Python, FastAPI, LangChain, ChromaDB, Gemini 3.5 Flash

- Engineered a modular Retrieval-Augmented Generation (RAG) pipeline utilizing FastAPI and LangChain to execute semantic context searches over unstructured PDF data.
- Integrated Google’s **gemini-embedding-001** and **gemini-3.5-flash** models to translate text into high-dimensional coordinates, producing context-grounded, cited answers.
- Designed a custom obsidian-themed Streamlit interface featuring dynamic text search and metadata-extracted source citations, deploying both client and backend servers asynchronously across Render and Streamlit Cloud.

PlacementTutor – Full-Stack Mentorship Booking Platform | [GitHub](#) | [Live](#)

Node.js, Express.js, MongoDB, Mongoose, JWT

- Developed a REST API (Express.js, Mongoose) with role-based access control across 3 user roles and dual authentication (email OTP + bcrypt), serving mentors, students, and admins.
- Designed an atomic reservation engine using MongoDB’s **findOneAndUpdate**, fully resolving concurrent double-booking race conditions at the database level.
- Integrated Razorpay payments with HMAC signature verification and a 5-minute payment-protected slot-lock using a lazy-cleanup pattern.
- Built a slot-splitting algorithm partitioning mentor availability into 15/30/60-min atomic segments, with per-user booking limits and cancellation logic.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, JavaScript, C++, SQL **AI/ML:** LangChain, RAG, LLM APIs, ChromaDB, Embeddings
Full-Stack: Node.js, Express.js, FastAPI, React.js, MongoDB, JWT **Tools:** Git, GitHub, Postman, Render
Areas of Interest: AI Product Engineering, Applied LLMs, Backend Development

POSITIONS OF RESPONSIBILITY

Publicity Executive, Blitzschlag (Annual Cultural Fest)	2025
MNIT Jaipur	
Conducted inter-college outreach and promotional tours, engaging with diverse student communities to maximize festival visibility and registration.	
Technical Executive, Sphinx (Technical Fest)	2025
MNIT Jaipur	
Facilitated technical workshops and outreach initiatives, interacting with technology enthusiasts to drive participation in technical competitions.	

ACHIEVEMENTS

Best Filo Teacher	October 2025
Filo Education	
Mentored a large student base with top-tier instructor ratings.	
First Prize – College Science Club Event	2024
MNIT Jaipur	
Secured first place in a highly competitive scientific contest.	